Nikita Kazeev

kazeevn@gmail.com · Website · GitHub · LinkedIn · Google Scholar · ORCID

Research Scientist focusing on Reliability and Uncertainty in Agentic Systems

Work Experience

Perfomax *AI Advisor* **2026–present**

Constructor Group *Machine Learning Consultant* **2021–present**

National University of Singapore *Postdoc under Kostya Novoselov* **2022–present**

Leadership

- Program Chair for the AI4X 2025/6 conferences
- Main organiser of the ICLR 2025 Workshop on Machine Learning Multiscale Processes
- Co-PI of a US$3.4m AI Singapore grant on multiscale machine learning

LLM Agent Safety & Alignment ICML 2026 Workshop · ICML 2026 Workshop · GitHub

- Conceived and developed a novel geometric framework to predict multi-step reasoning failures in LLM agents using semantic abstraction trajectories
- Designed uncertainty quantification (UQ) algorithms for LLM reasoning loops, improving agent error detection and calibration
- Developed and open-sourced reference-checker, a tool for verifying claims and citation references in scientific papers to detect and mitigate LLM hallucinations
- Authored a position paper on Pluralistic Alignment, advocating for aligning AI models with aspirational human values rather than replicating behavioral flaws

Wyckoff Transformer, a generative model for materials ICML 2025 · GitHub

- Alignment by design: enforced physical principles directly at the architectural level as hard constraints, mathematically guaranteeing plausible generations
- Solved a systemic failure mode where generative models default to asymmetric outputs despite symmetric training data, mitigating this structural bias via coordinate-free representations
- Increased materials discovery yield (S.U.N.) by 1.24x while cutting inference time by 4 orders of magnitude (to 0.05 GPU-ms per structure)
- Orchestrated large-scale (10k structures) DFT computation loops to verify out-of-distribution stability; led a team of 6

⟨CERN | Yandex → HSE University⟩ *Researcher* **2014–2022**

2025 Breakthrough Prize in Fundamental Physics as a member of LHCb.

Generative models uncertainty estimation paper · conference · code 1 · code 2

- Developed the first methods for estimating uncertainty of conditional GANs
- Led a team of 3 students

Machine learning on noisy data paper 1 · paper 2 · code

- Derived a statistically rigorous method to train classifiers on sPlot-weighted (label-noisy) data, standard in high-energy physics

CatBoost: Combatting Biases in Boosting paper

- Designed and built distributed experiment infrastructure to scale systematic bias analysis across large-scale tabular datasets
- Investigated dynamic boosting formulations to eliminate target leakage and prediction shift, mitigating statistical bias in gradient boosting

Education

HSE University & Sapienza Università di Roma **2016–2020**
PhD in Computer Science and Physics (Double Degree)
Supervisors: Andrey Ustyuzhanin & Barbara Sciascia
Thesis: Machine Learning for particle identification in the LHCb detector

Product Management course at Yandex **2018**
Focused coursework in product strategy and execution.

Yandex School of Data Analysis **2013–2015**
Master's-level CS programme
Algorithms, machine learning, deep learning, and distributed systems.

Moscow Institute of Physics and Technology
MS in Physics **2014–2016**
Optimisation of data processing of the LHCb experiment.
BS in Physics **2010–2014**
Wave-packet molecular dynamics of electrons in nonideal plasma.

International Junior Science Olympiad (IJSO) **Korea, 2008**
Silver medal

Mentorship

- Mentored 9 students (1, 2, 3, 4, 5, 6, 7, 8, 9), 3 interns, and student workshop projects (1, 2).

Teaching & Outreach

- Pitch research to the general public through major newspaper coverage (1, 2), blog posts (1, 2), a science slam, interviews (1, 2), and a public talk.
- Lecturer in the award-winning Coursera Advanced Machine Learning specialisation
- Delivered more than 20 conference talks.

Service

- Reviewer for NeurIPS, RSC Advances, Machine Learning: Science and Technology, Digital Discovery
- Peer Staff Supporter at NUS, serving as a first-line support for mental wellbeing.

Skills

- **Agent Reliability & UQ** — developed geometric and statistical frameworks to predict LLM agent reasoning failures using semantic abstraction trajectories.
- **Aspirational Alignment & Verification** — formulated alignment strategies based on high-level human values and established benchmarks (e.g. Foresight-Phys) for validating AI systems in complex environments.
- **Uncertainty quantification** — built the first methods for estimating the uncertainty of conditional GANs; calibrated generative and discriminative models.
- **Inference and systems at scale** — C++ production integration, large-scale DFT/VASP on HPC clusters, and reproducible experiment pipelines.
- **Research leadership** — lead multidisciplinary teams, chair conferences, and mentor students from idea to publication.

ML & AI • Python • PyTorch • C++ • HPC • Physics • Research Writing & Speaking • Leadership & Mentorship

Presenting at ICML 2026

July 11, 12:15–13:30 — **Nikita Kazeev** and Ian Babich. "Foresight-Phys: A Benchmark for Forecasting the Results of Physical Experiments". In: *Forecasting as a New Frontier of Intelligence Workshop at ICML 2026*.

July 11, 15:00–16:30 — **Nikita Kazeev** and Phan Bui Nhat Huyen. "Position: Align AI to Our Aspirations, Not Our Flaws". In: *Pluralistic Alignment Workshop at ICML 2026*.

July 11, 15:30–17:00 — **Nikita Kazeev** and Andrey Ustyuzhanin. "The Geometry of Reasoning Failure: Predicting Agent Errors from Semantic Scale Trajectories". In: *Statistical Frameworks for Uncertainty in Agentic Systems Workshop at ICML 2026*.

July 10, 11:40–13:30 & 16:10–17:00 — **Nikita Kazeev** and Andrey Ustyuzhanin. "Uncertainty Quantification for LLM Agents via Semantic Abstraction Trajectories". In: *Structured Probabilistic Inference & Generative Modeling Workshop at ICML 2026*.

Selected Publications

Nikita Kazeev, Wei Nong, Ignat Romanov, Ruiming Zhu, Andrey E Ustyuzhanin, Shuya Yamazaki, and Kedar Hippalgaonkar. "Wyckoff Transformer: Generation of Symmetric Crystals". In: *Proceedings of the 42nd International Conference on Machine Learning*. Ed. by Aarti Singh, Maryam Fazel, Daniel Hsu, Simon Lacoste-Julien, Felix Berkenkamp, Tegan Maharaj, Kiri Wagstaff, and Jerry Zhu. Vol. 267. Proceedings of Machine Learning Research. PMLR, 13–19 Jul 2025, pp. 29495–29526.

Maxim Borisyak and **Nikita Kazeev**. "Machine Learning on data with sPlot background subtraction". In: *Journal of Instrumentation* 14.08 (2019). **Alphabetic order, I'm the corresponding author.**

Denis Derkach, **Nikita Kazeev**, Fedor Ratnikov, Andrey Ustyuzhanin, and Alexandra Volokhova. "Cherenkov detectors fast simulation using neural networks". In: *Nuclear Instruments and Methods in Physics Research Section A* (2019). **Alphabetic order, I'm the corresponding author.**